RIGIDITY MODELING THEORY AND PHYSIOLOGIC INFERENCE FRAMEWORK

Concept of Inferred Rigidity

Penile rigidity may be modeled as an inferred physiologic condition derived from correlated mechanical, vascular, thermal, and motion-related measurements rather than direct measurement of internal pressure or anatomical stiffness. The wearable device may evaluate deformation behavior, tissue interaction forces, displacement characteristics, and dynamic response patterns to estimate functional rigidity states across time.

Rigidity inference may be based on relationships between circumferential expansion, resistance to deformation, rate of dimensional change, pulsatile mechanical behavior, and stability of maintained structural configuration. Computational models may interpret these relationships to produce a continuous rigidity index or categorized physiologic state representation.

Mechanical Modeling Approaches

Mechanical interpretation may incorporate elastic deformation modeling, compliance estimation, force-displacement relationships, and time-dependent stiffness response analysis. Device measurements may approximate tissue rigidity by observing structural reaction to physiologic expansion rather than requiring invasive pressure sensing. Modeling techniques may include linear, nonlinear, empirical, or machine-learned mechanical response frameworks.

Mechanical models may account for anatomical variability, device fit characteristics, sensor placement differences, and individual tissue response profiles. Calibration procedures may establish individualized reference baselines enabling normalization of rigidity estimates across users and over longitudinal monitoring periods.

Dynamic State Transition Modeling

Physiologic state transitions such as tumescence onset, rigidity stabilization, detumescence, and recovery may be modeled as temporal processes characterized by rate-of-change parameters. Analysis may evaluate slopes, accelerations, inflection points, oscillatory behavior, and episode continuity to classify dynamic physiologic responses.

Temporal modeling may enable differentiation between rapid activation events, gradual transitions, interrupted episodes, and sustained physiologic states. Time-domain analysis may be combined with contextual sensing information to improve interpretation confidence.

Multi-Modal Sensor Fusion Framework

Rigidity estimation may incorporate sensor fusion techniques combining strain, pressure, magnetic displacement, temperature, optical perfusion, and inertial measurements. Fusion algorithms may apply weighted averaging, probabilistic inference, statistical modeling, rule-based logic, or machine learning methods to integrate heterogeneous sensor inputs into coherent physiologic interpretations.

Sensor fusion may increase robustness against noise, motion artifacts, environmental variation, and anatomical diversity by relying on agreement between independent sensing channels rather than dependence upon a single measurement modality.

Longitudinal Learning and Adaptive Modeling

Over extended monitoring periods, computational models may adapt to user-specific physiologic patterns through longitudinal learning processes. Adaptive models may refine rigidity estimation parameters, improve prediction accuracy, and personalize interpretation thresholds based on historical data trends.

The modeling concepts described herein are illustrative and non-limiting and encompass present and future computational methods capable of inferring physiologic rigidity and state transitions from multi-modal wearable sensing data.